

Program : Diploma in Instrumentation Engineering	
Course Code : 3084	Course Title: Control Engineering
Semester : 3	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- To provide a thorough knowledge about control engineering concepts.
- To provide an ability to comment on the stability of systems if the transfer functions are given.
- To lay foundation for further courses like Process Control Instrumentation, Industrial automation etc.

Course Prerequisites:

Topic	Course Code	Course name	Semester
Fundamentals of Electrical & Electronics		Fundamentals of Electrical & Electronics	2
Force and Torque		Applied Physics II	2
Differentiation and Integration		Mathematics II	2

Course Outcomes:

On completion of the course, students will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Represent linear time invariant system using Laplace transform	15	Applying
CO2	Derive transfer functions of simple systems.	15	Applying
CO3	Analyze the time domain and frequency domain characteristics of an LTI system.	14	Applying

CO4	Analyze the stability of an LTI system using Routh Hurwitz criterion, Root locus and Bode Plot.	14	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Represent linear time invariant system using Laplace transform		
M1.01	Explain the concepts of control system and mathematical modeling	4	Understanding
M1.02	Compare linear time invariant and linear time variant system.	3	Understanding
M1.03	Apply the concept of Laplace transform, inverse Laplace transform and theorems of Laplace transforms to solve problems	4	Applying
M1.04	Solve differential equations using Laplace transform, inverse Laplace transform and theorems of Laplace transforms	4	Applying
Contents: Basics of control system - physical model - mathematical model of control system - linear time invariant and linear time variant system - open loop and closed loop control systems - Laplace transforms - Laplace transform of e^{at} , e^{-at} , t , $\sin(\omega t)$ and $\cos(\omega t)$ - Laplace transform of step, ramp, parabolic, impulse inputs - Laplace transform theorems – Real differentiation theorem, Real integration theorem (proofs not required), Initial Value theorem and final value theorem (derivation of theorems not required) - Inverse Laplace transforms - partial fraction method to solve simple problems (up to 2nd order systems).			

CO2	Derive transfer functions of simple systems.		
M2.01	Explain the concept of transfer function	3	Understanding
M2.02	Derive the transfer function of electrical and mechanical systems.	4	Applying
M2.03	Describe the analogy of electrical and mechanical system parameters.	3	Understanding
M2.04	Derive the overall transfer function of systems using signal flow graph and block diagram reduction methods.	5	Applying
	Series Test – I	1	
Contents: Transfer function- definition and order - transfer function of linear system - general transfer function of Mechanical Translational system and rotational system - transfer function of Electrical circuits – R-L-C (series & parallel) - force/torque - voltage and force/torque - current analogy. Block diagram reduction - rules - overall transfer function of control systems using block diagram reduction rules. Signal flow graph(SFG)-terms - Mason's gain formula - overall transmittance of control system by signal flow graph.			
CO3	Analyze the time domain and frequency domain characteristics of an LTI system.		
M3.01	Determine the time response of first and second order systems for standard test signals	5	Applying
M3.02	Derive error coefficient of static position, velocity & acceleration	4	Applying
M3.03	Derive the steady state error of Type 0, Type 1 and Type 2 Systems	5	Applying
Contents: Time response analysis - time response of first order system for step, ramp, and impulse inputs, time constant-time response of second order system for step input, damping ratio, natural frequency, damped frequency, over damped, critically damped and under damped systems - type of a system - static error coefficients - static position, velocity and acceleration error coefficient - steady state error in terms of Kp, Kv and Ka for Type 0, Type 1, Type 2 Systems			
CO4	Analyze the stability of an LTI system using Routh Hurwitz criterion, Root locus and Bode Plot.		
M4.01	Explain absolute stability, relative stability and marginal stability	3	Understanding
M4.02	Analyze the stability of a system by using Routh Hurwitz criterion	4	Applying

M4.03	Analyze the stability of given transfer function of a system using Bode plot	4	Applying
M4.04	Construct root locus for the given transfer function	3	Applying
	Series Test – II	1	
Contents: Absolute stability, relative stability, marginal stability- Routh Hurwitz criterion, stability analysis of systems using Routh Hurwitz criterion (order up to six), Bode plot – stability analysis of simple transfer functions. - gain cross over frequency, phase cross over frequency, gain margin, phase margin (limited to K , K_s , K/s , $1+Ts$, $1-Ts$, $1/1+Ts$, $1/1-Ts$ only) - Root locus - Procedure to construct Root Locus.			

Text / Reference:

T/R	Book Title/Author
R1	I.J. Nagarath, N. Gopal , Control systems Engg. , New Age International Publishers
R2	R.S. Manke ,Linear Control Systems , Khanna Publisher
R3	Katsuhiko Ogata, Modern Control Engineering, Khanna Publisher
R4	R. Anandanatarajan.,P. Ramesh Babu, Control Systems Engineering, Scitech. Publisher
R5	A Nagoor Kani, Control System Engineering, RBA Publications

Online Resources:

Sl. No	Website Link
1	https://www.tutorialspoint.com/control_systems/index.htm
2	https://www.electronics-tutorial.net/control-systems/
3	https://www.youtube.com/user/ControlLectures
4	https://www.electrical4u.com/control-system-closed-loop-open-loop-control-system/